Summary in Graph

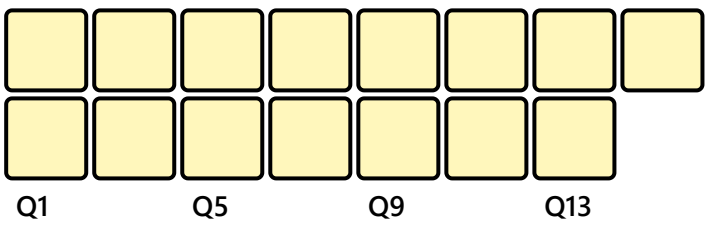
Exam Summary (GO Classes CS/DA | Conditional Probability | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (8 + 7)
Total Marks:	22 (8 + 14)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



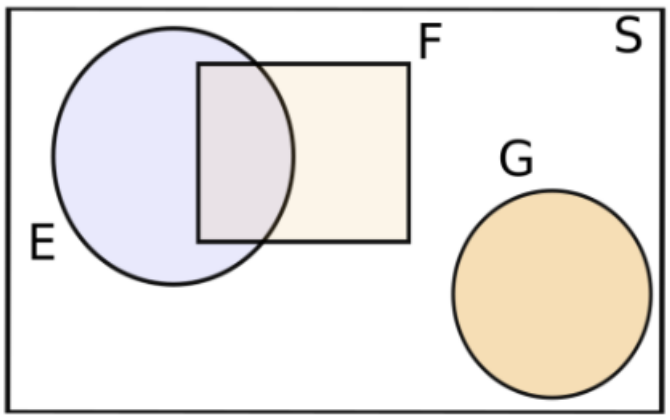
* Average Stats based on first-attempt stats of Top 10% (75) out of 752 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Probability

Which all of the following is true for the Venn diagram shown here:



A. $P(E \cup G) = P(E) + P(G)$
B. $P(E \cup G) = P(E) + P(G) - P(E \cap G)$
C. $P\left(E^C \cap F\right) = P\left(F^C\right) - P(E)$
D. $P\left(E^C \cap F\right) = 1 - \left(P\left(F^C\right) - P(E)\right)$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following is/are CORRECT?

A. If events E_1 and E_2 are statistically independent, then

$$\mathrm{P}\left(E_1 \cup E_2\right) = \mathrm{P}\left(E_1\right) + \mathrm{P}\left(E_2\right)$$

B. If events E_1 and E_2 are mutually exclusive, then

$$\mathrm{P}\left(E_1 \mid E_2\right) = 0$$

C. If E_1, E_2, and E_3 are mutually exclusive and collectively exhaustive, then

$$\mathrm{P}\left(E_1\right) = 1 - \mathrm{P}\left(E_2\right) - \mathrm{P}\left(E_3\right)$$

D. $\mathrm{P}\left(\overline{E_1} \mid E_2 \cup E_3\right) = 1 - \mathrm{P}\left(E_1 \mid \overline{E_2 \cup E_3}\right)$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Suppose that k events $B_1, B_2, \ldots, B_k$ form a partition of the sample space S. There is another event A that $\Pr(A) > 0$.

Which of the following is/are CORRECT?

A. $\sum_{i=1}^{k} P\left(B_i\right) = 1$
B. $\sum_{i=1}^{k} P\left(B_i \mid A\right) = 1$
C. $\sum_{i=1}^{k} P\left(A \mid B_i\right) = 1$
D. $\sum_{i=1}^{k} P\left(B_i \wedge A\right) = P(A)$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, what must be true:

A. $P(A \mid B)$ decreases
B. $P(B \mid A)$ decreases
C. $P(B)$ decreases
D. All of above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Alice, Bob, and Carl each attempt to solve a crossword puzzle. There is a 70% chance that Alice can solve the puzzle without making a mistake, a 60% chance that Bob can, and a 85% chance that Carl can. What is the probability that each one makes a mistake in solving the puzzle?

A. 0.018
B. 0.357
C. 0.9
D. 0.12

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following(s) expression is/are the same as $P(A, B, C)$ given no independent assumptions?

A. $P(C \mid A, B) \cdot P(A, B)$
B. $P(C \mid A, B) \cdot P(A) \cdot P(B)$
C. $P(A \mid B) \cdot P(B \mid C) \cdot P(C)$
D. $P(A \mid B, C) \cdot P(B \mid C) \cdot P(C)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Suppose we wish to calculate $P(H \mid E1, E2)$, and we have no conditional independence information.

Which of the following sets of numbers are sufficient for the calculations?

A. $P(E1, E2), P(H), P(E1 \mid H), P(E2 \mid H)$
B. $P(E1, E2), P(H), P(E1, E2 \mid H)$
C. $P(H), P(E1 \mid H), P(E2 \mid H)$
D. $P(H), P(E1 \mid H)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Suppose A and B are independent events. Consider the following probabilities.

$$P(A, B) = 0.15$$
$$P(A, B^c) = 0.45$$
$$P(A^c, B) = x$$
$$P(A^c, B^c) = y$$

What will be the value of $x + y$?

Your Answer: | Correct Answer: 0.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Which of the following statements is NOT true?

A. If $A \subset B$, then $P(A) \leq P(B)$.
B. If $P(B) > 0$, then $P(A \mid B) \geq P(A)$.
C. $P(A \cap B) \geq P(A) + P(B) - 1$.
D. $P(A \cap B^c) = P(A \cup B) - P(B)$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

I have two six-sided dice. The first die has 3 faces painted red and the other 3 painted black. The second has 1 red face and 5 black faces. When I roll a die, each of the six faces are equally likely. I choose a die at random (both dice being equally likely), and roll it twice (rolling the same die both times).

What is the conditional probability that I chose the die with 3 red faces, given that the first roll came up "red"?

A. $1/2$
B. $3/4$
C. $1/6$
D. $1/3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A drug-screening test is used in a group of professional baseball players of whom 4% actually use illegal drugs. It is found that the test indicates positive in 97% of those who use drugs and 2% of those who do not.

What is the probability that a randomly chosen player in the group with positive test result actually uses drugs?

A. 0.67
B. 0.97
C. 0.06
D. 0.52

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

- Class has 50 students
- 20 male (M), 25 brown-eyed (B)

For a randomly chosen student, what is the range of possible values for $p = P(M \cup B)$?

A. $p \leq .4$
B. $.4 \leq p \leq .5$
C. $.4 \leq p \leq .9$
D. $.5 \leq p \leq .9$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A box contains three coins: two regular coins and one fake two-headed coin $(P(H) = 1)$, You pick a coin at random and toss it.

What is the probability that it lands heads up?

A. $1/2$
B. $2/3$
C. $4/5$
D. None of these

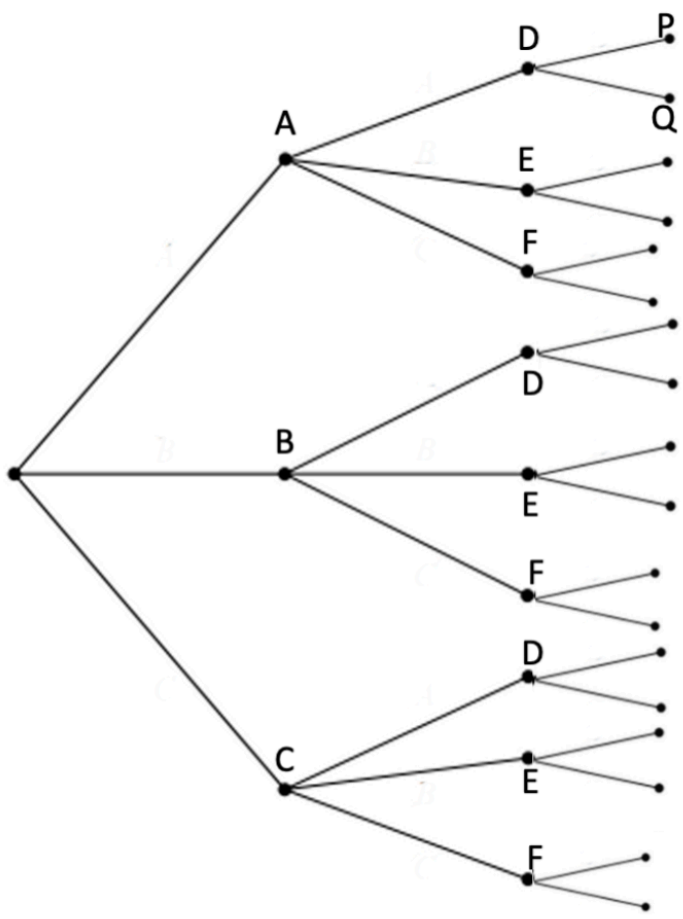
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a three level tree diagram of events happening in a sequence.

At first level A, B or C events are possible, at second level D, E or F are possible and at third level P or Q are possible.

All edge probabalties at first level are $1/3$, at second level are $1/3$, and at third level are $1/2$.



Find $P(A \mid P)$?

A. $2/3$
B. $1/3$
C. $1/2$
D. $3/4$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

For a sample space S encompassing three events E, F, and G, if it is given that $P(E) = 0.3, P(F) = 0.8, P(G) = 0.2$, which of the following can NOT be true?

A. Events E and F are mutually exclusive events
B. Events F and G are mutually exclusive events
C. Events E and F are independent events
D. Events F and G are independent events

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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